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UGC AUTONOMOUS INSTITUTE

Design and Implementation of Automated Answer Evaluation System

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Group no : **A8**

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Introduction

Abstract

- Problem: Manual grading of handwritten descriptive answers is time-consuming and inconsistent.
- Gap: Existing tools mainly support objective questions or require strict formatting.
- Solution: An AI system that uses semantic analysis (Sentence-BERT) to evaluate.
- Functionality: Compares extracted text from student answers with model answers based on meaning, not exact words.
- Benefits: Reduces human effort, improves grading fairness, and is easy for educators to use with minimal setup.

Problem Definition

To design, develop and implement artificial intelligence based answer sheet evaluator

Background

Manual evaluation of descriptive answers is time-consuming and error-prone. It often results in inconsistent grading and increased workload for educators. Most existing automated tools focus on objective-type questions or fixed formats. They lack support for free-form, descriptive responses. Recent AI advancements in NLP enable smarter text understanding. Semantic analysis models like Sentence-BERT allow meaning-based answer comparison. This project proposes an AI system to evaluate handwritten answers fairly and efficiently. It aims to reduce manual effort while ensuring accurate and consistent grading.

Aim

To design, develop, and implement an AI-based system that automatically evaluates descriptive answer sheets using semantic analysis, ensuring fair, consistent, and efficient grading with minimal human intervention.

Objectives

- To provide constructive feedback that highlights strengths and suggests improvements for learners.
- To design an automated system capable of evaluating descriptive student answers with consistency and fairness.
- To integrate semantic similarity techniques for assessing the conceptual alignment between student and reference answers.
- To implement a mechanism for identifying irrelevant or off-topic content in responses.

Prior Art (Patent)

Automated answers evaluation and scoring system and method. *US5180309A*

It describes a system for automated evaluation and scoring of typed or handwritten answers. It uses pattern matching, redefined answer sets, and a randomized question delivery mechanism. The system assigns full or partial credit based on tolerance levels.

Semi-automated evaluation of long answer exams
US11967253B2

The system uses OCR to extract handwritten answers, applies NLP and
machine learning for content evaluation, and combines a confidence-based scoring engine with a human-in-the-loop for grade review and adjustment.

System and methods for distilled BERT based training model for text classification. *US11922303B2*

It describes a method to create a smaller, faster BERT model using knowledge distillation from a larger BERT. The resulting model performs well in text classification tasks while being more efficient.

Answer Evaluation Method, Answer Evaluation System, Electronic Device, and Medium .*US11790641B2*

It uses neural networks combined with OCR to detect different test paper types and accurately recognize both text and mathematical symbols. It divides scans into objective and subjective regions and supports automated marking across question types.

Literature Review

Short Answer Grading Using One shot Prompting and Text Similarity Scoring Model (May 2023)

Proposed by Su-Youn Yoon

This paper introduces a hybrid method for grading short answers using one-shot prompting with large language models (LLMs) and semantic similarity scoring. The approach enables effective evaluation with minimal training data by combining prompt-based response generation and embedding-based similarity comparison.

Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (August 2019)

Proposed by Nils Reimers, Iryna Gurevych.

Introduces SBERT, which fine-tunes BERT with Siamese training to yield sentence embeddings suitable for fast semantic similarity evaluation—ideal for answer grading.

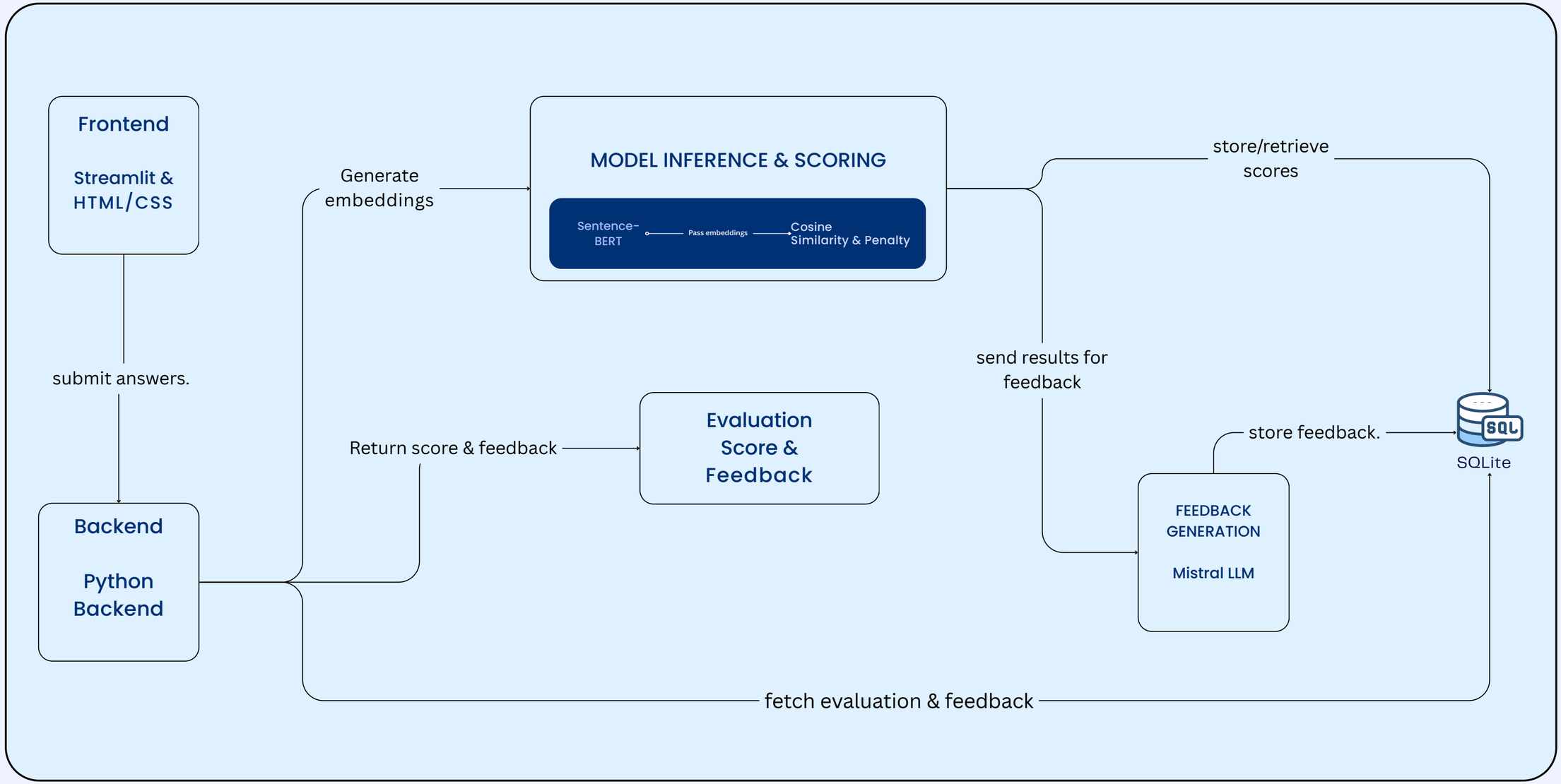
Literature Review

Beyond Human Subjectivity and Error: A Novel AI Grading System (May 2024)
Proposed by Alexandra Gobrecht, Felix Tuma, Moritz Möller, et al.

This paper proposes an AI grading system using fine-tuned transformer models for short answer evaluation. Trained on university exam data, it outperforms human graders in consistency and accuracy. The approach combines deep learning and NLP to reduce subjectivity and improve fairness in assessments.

Automated Content Grading Using Machine Learning (April 2020)
Proposed by Rahul Kr Chauhan, Ankit Kumar, Baisakhi Chakraborty, Hiteshwar Kumar.
Chauhan et al. (2020) applied machine learning techniques like bag-of-words and TF-IDF to evaluate student responses against model answers. Their approach combines lexical features and semantic similarity to achieve reliable automated grading, making it a practical solution for descriptive assessments.

Proposed Approach and System Architecture



Proposed Solution

- The system accepts student answers in text format as input.
- It employs SBERT to compute the semantic similarity between the student's response and the reference answer.
- A penalty is applied if the computed similarity score falls below a threshold.
- An LLM-based feedback generator provides constructive suggestions to improve the answer.
- The final output includes the similarity score, the penalty-adjusted score, and detailed feedback.

Key Benefits

- Provides consistent evaluation of descriptive answers.
- Reduces manual grading workload for educators.
- Delivers constructive, LLM-generated feedback to improve learning outcomes.
- Ensures fair scoring through semantic similarity analysis with penalty thresholds.

System Workflow and Implementation

Input Collection

For a question, model and student answer are taken as input.

(From a simple Web Interface)

Sentence Segment

Divide answer into sentence.

(Using Regular Expression in Python)

Tokenization

Convert sentence to tokens.

(Hugging Face Model)

Vector Embedding generation

Creation of embedding using sentence-BERT.

(Sentence-BERT)

Embedding Comparison

Compare embedding of model and student answer using cosine similarity.

(Using Cosine Similarity)

Penalty Application

Apply penalties for low similarity.

(Python)

Feedback Generation

Generate a feedback on student answer using LLM.

(Large Language Model)

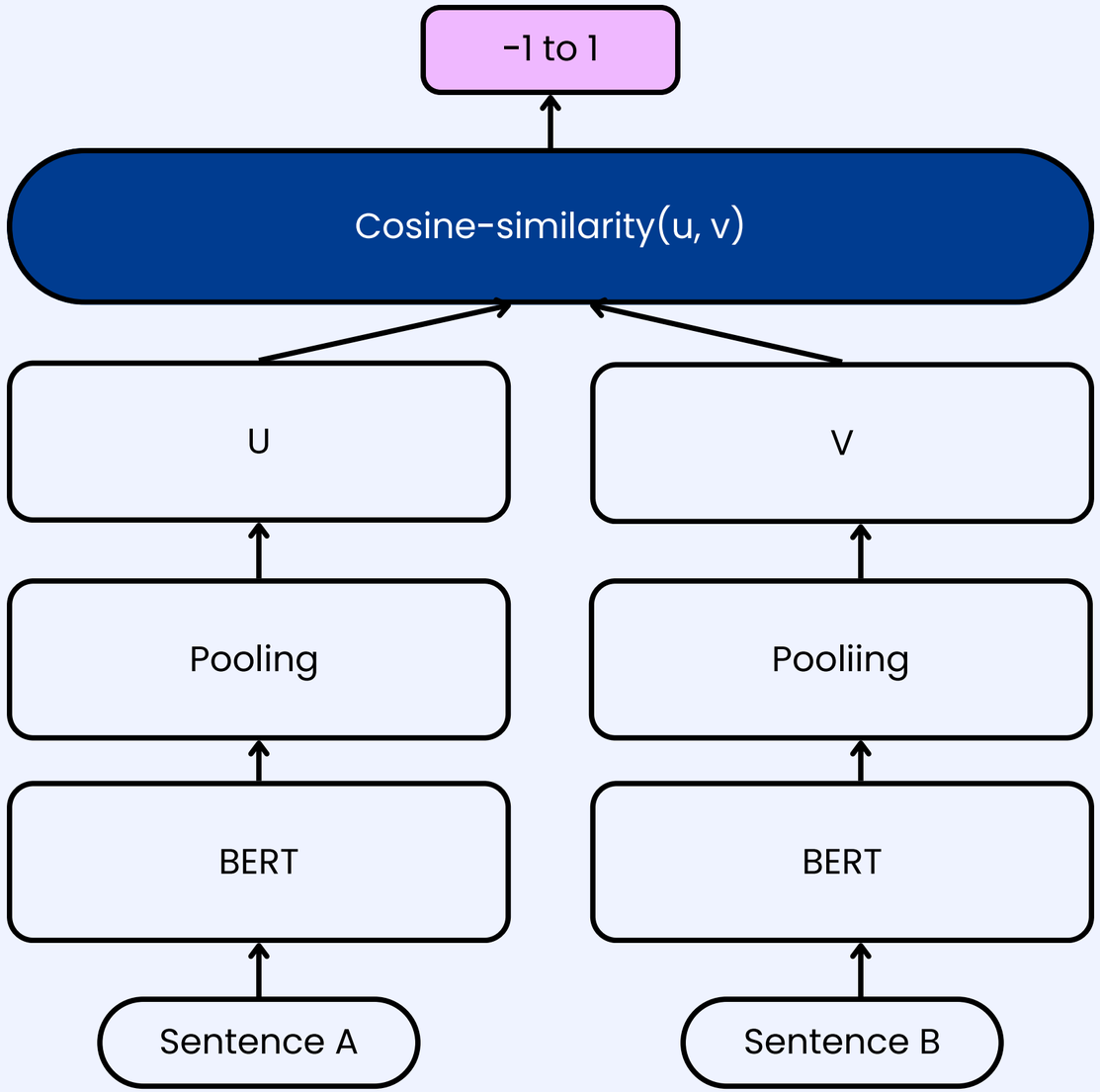
Final Score Calculation

Compute the final score

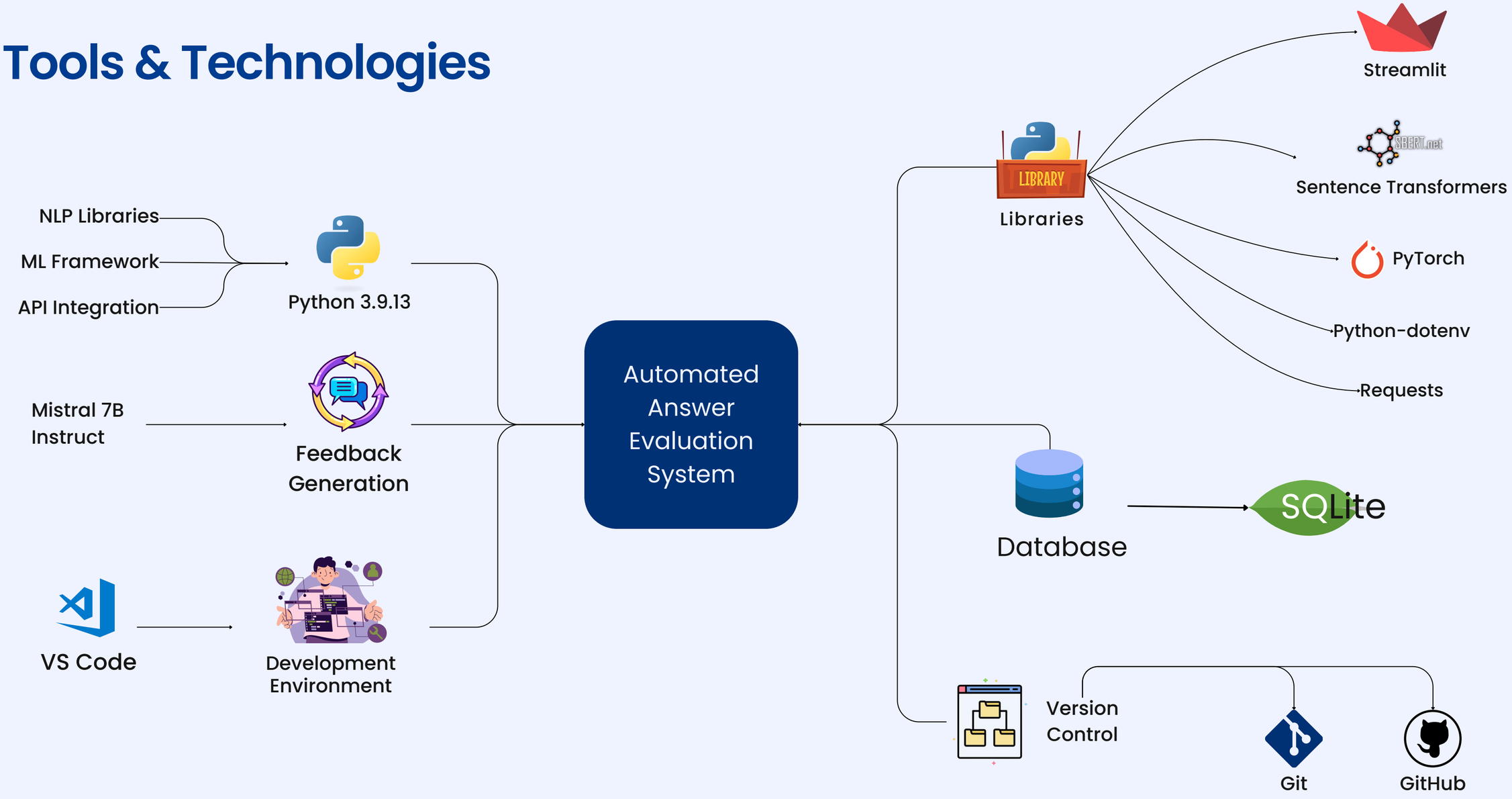
(Python)

Sample Code

```
from sentence_transformers import
SentenceTransformer, util

model = SentenceTransformer("all-MiniLM-L6-v2")
sentences = ["Apple unveiled the new phone.",
"Please buy an apple and an Orange."]

embeddings = model.encode(
sentences, convert_to_tensor=True)

similarity = util.cos_sim(
embeddings[0], embeddings[1])

print(f"Cosine Similarity: {similarity.item():.4f}")
```



Tools & Technologies



Result

- Student submissions stored successfully in SQLite.
- Semantic similarity computed with SBERT and cosine similarity
- Rubric scores generated using Mistral 7B.
- Final score combined similarity and rubric evaluation.
- Teacher overrides allowed manual flexibility.
- Three answer types tested: correct, partial, irrelevant.
- Correct answers scored high and matched reference well.
- Partial answers received moderate scores with feedback.
- Irrelevant answers were penalized effectively.
- System showed consistency, fairness, and robustness.

Output Sceenshots

View Student Results

Download All Results (CSV)

Select Student

01 – Charlie

Student: Charlie

Roll Number: `01`

Download Charlie's Marksheet (CSV)

What is a compiler?

Submitted on: 2025-09-11T16:03:58.514686

Student Answer

`A compiler is software that translates high-level language programs into machine code so the computer can execute them. It performs several phases such as lexical analysis, syntax analysis, semantic checks, optimization, and code generation.` These steps ensure correctness of the code, handle errors, and make the final program efficient to run.

Similarity Score: `0.8803`

Final Score (AI): `0.9162`

Rubric Scores:

- Factual Accuracy: 5/5
- Completeness: 5/5
- Clarity: 5/5
- Relevance: 5/5

Feedback:

Excellent job! Your answer accurately and clearly explains what a compiler is and the steps it takes. You also provide a concise summary of the functions of each phase. Keep up the good work!

Teacher Override Score

0.92

Save Override

View Student Results

Download All Results (CSV)

Select Student

03 – Dunno

Student: Dunno

Roll Number: `03`

Download Dunno's Marksheet (CSV)

What is a compiler?

Submitted on: 2025-09-11T16:05:13.136850

Student Answer

A compiler is basically like an operating system because it manages the hardware of the computer. It is responsible for controlling files, memory, and applications that help the computer start and function. Without it, the computer would not be able to boot up or handle user tasks.

Similarity Score: `0.5261`

Final Score (AI): `0.4583`

Rubric Scores:

- Factual Accuracy: 1/5
- Completeness: 2/5
- Clarity: 2/5
- Relevance: 1/5

Feedback:

The response provides some correct information about the role of a compiler, but it is not accurate to compare a compiler to an operating system. The response is incomplete as it does not mention the translation of high-level programming languages into machine code. The response is not clear in its explanation of how a compiler manages hardware or controls files, memory, and applications. The response is not relevant as it does not discuss the phases of compilation or the error checking and optimization processes.

Teacher Override Score

0.46

Save Override

Future Scope

- Incorporate and fine-tune multiple pre-trained models tailored to specific subjects or domains to improve semantic understanding and grading reliability.
- Develop mechanisms to detect irrelevant or off-topic responses, ensuring fairer and more accurate scoring.
- Evolution into a fully automated and intelligent grading system.
- Capability to process diagrams and mathematical equations in student responses.
- Support for varied file type inputs such as PDF, DOCX, and images.

Conclusion

This project presents an Automated Answer Evaluation System that uses advanced NLP models to address the limitations of manual and keyword-based grading. By combining Sentence-BERT for semantic similarity, Mistral 7B for rubric-based assessment, and a structured SQLite backend, it ensures fairness, consistency, and transparency in evaluating descriptive answers. With its modular design, Streamlit interface, and feedback generation, the system reduces evaluator workload while supporting student learning through constructive insights.

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Thank You